

7 The curve C has polar equation $r^2 = (\pi - \theta) \tan^{-1}(\pi - \theta)$, for $0 \leq \theta \leq \pi$.

(a) Sketch C and state the polar coordinates of the point of C furthest from the pole. [3]

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(b) Using the substitution $u = \pi - \theta$, or otherwise, find the area of the region enclosed by C and the initial line. [7]

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